Heat, Mortality, and Life-Cycle Welfare

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October 27, 2026

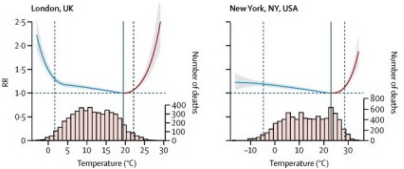
Motivation



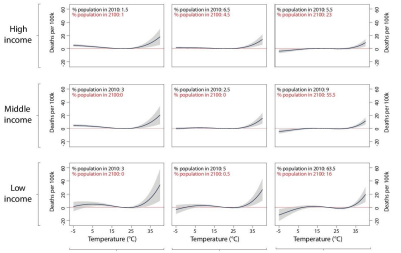
“Heat is the deadliest form of extreme weather.”

1 July 2024

The Washington Post



Relative risk vs. temperature (°C). Source: Gasparrini et al. (2015).



Source: Carleton et al. (2022).

This Paper

Stochastic life-cycle model of heat-induced mortality risk, calibrated to U.S.

- **High-frequency damage:** convex mortality temperature nexus ⇒ frequency *and* intensity of hot days matter
- **Private adaptation:** costly protection (AC, shading, relocation) dampens heat-mortality risk ⇒ protection rises with income
- **Individual heterogeneity:** welfare effects depend on ...
  - ▷ exogenous: earnings, chronological age, expectations about heat
  - ▷ endogenous: adaptation, biological age, savings
- **Outcomes:** ΔT and ΔLTU

⇒ Structural framework mapping daily heat shocks into individual outcomes

Method

- Stochastic life-cycle model
  - Uncertain survival (Yaari, 1965)
  - Endogenous health (Dalgaard and Strulik, 2014)
  - Daily heat shocks and private adaptation
- Bringing the model to the data
  - Parameters estimated by SMM
  - Heat shocks: daily mean temperature distribution
  - SSP2-4.5, SSP3-7.0, SSP5-8.5 against an SSP1-2.6 benchmark
  - Solving mixed-frequency model using relaxation algorithm
- Analytical reduced-form model to structure quantitative results

Stylized Model

Constant mortality. Log utility, $c = w > 1$, $u(\text{death}) = 0$. Burden of m?

$$\underbrace{\ln\big((1-\lambda)\,w\big)}_{\text{baseline}} = \underbrace{(1-m)\ln w}_{\text{alternative}} \quad \Rightarrow \quad \lambda = 1 - e^{-\Lambda(w)}, \qquad \Lambda(w) \equiv m \ln w$$

$$\Rightarrow \quad \text{Given } \varepsilon_{\text{VSL}} > 1, \text{ burden is progressive: } \Lambda'(w) = \frac{m}{w} > 0 \quad \Leftrightarrow \quad \partial\lambda/\partial w > 0$$

Income-dependent mortality. Assume $m = m(w)$ with $m'(w) < 0$.

$$\Lambda'(w) = \underbrace{\frac{m(w)}{w}}_{\substack{\text{utility scale} \\ (+)}} + \underbrace{m'(w)\ln w}_{\substack{\text{protection} \\ (-)}}$$

$\Rightarrow$ Two opposing forces with threshold w^*: progressive below it; regressive above

Research Questions and Major Results

- **Q1: How large are losses from heat-induced mortality risk?**
  - ΔT: 0.07 years / 0.8 months ΔLTU: 0.80% in CEV
- **Q2: How do losses vary along the income distribution?**
  - ΔT: 3.1 months (poor) vs. 0.5 months (rich)
  - ΔLTU: 1.32% (poor) vs. 0.52% (rich)
- **Q3: What mechanisms shape this incidence?**
  - Poor: survival channel; rich: consumption channel
  - Burden falls disproportionately on older agents
  - Climate denial hurts predominantly the rich

SSP5-8.5 vs. SSP1-2.6. ΔT: life expectancy; ΔLTU: lifetime utility (consumption equivalent).

Literature

Climate change impacts and inequality
Bilal and Känzig (2026), Bilal et al. (2026), Burke et al. (2015), Dell et al. (2012), Kotz et al. (2024), Hänsel et al. (2022)

Economics of adaptation
Carleton et al. (2024), Barreca et al. (2016), Fouillet et al. (2008), Carleton et al. (2022), Fried (2026), Fried (2022)

Climate epidemiology
Gasparrini et al. (2015), Vicedo-Cabrera et al. (2021), Masselot et al. (2025)

Gerontology and life-cycle theory
Mitnitski et al. (2001), Dalgaard and Strulik (2014), Yaari (1965), Hall and Jones (2007), De Nardi et al. (2010)

Stochastic Life-Cycle Model: Daily Heat and Adaptation

Life-Cycle Model: Basic Setup

$$\max_{\{c_t,\, k_t\}\geq 0} \sum_{t=0}^{\bar{T}} S_t\, \beta^t \underbrace{\left(\frac{c_t^{1-\sigma}-1}{1-\sigma} + \bar{u} \right)}_{\text{annual utility}} \quad \text{s. t.}$$

$$w_t + R_{t-1}\, k_{t-1} = c_t + k_t$$

$$\underbrace{S_t}_{\text{cum. survival}} = \prod_{v=0}^{t} \exp\Big(- \underbrace{m_v}_{\text{hazard}} \Big)$$

Yaari (1965)

Endogenous
c consumption
k wealth
S cumulative survival

Exogenous
w income, R interest
m_v age-specific hazard

Parameters
$\bar{T}$ max. lifetime; β, σ, $\bar{u}$

Life-Cycle Model: Biological Aging

$$\max_{\{h_t,\, c_t,\, k_t\} \geq 0} \sum_{t=0}^{\bar{T}} S_t \; \beta^t \underbrace{\left(\frac{c_t^{1-\sigma} - 1}{1-\sigma} + \bar{u} \right)}_{\text{annual utility}} \quad \text{s. t.}$$

$$w_t + R_{t-1}\, k_{t-1} = c_t + h_t + k_t$$

$$\underbrace{S_t}_{\text{cum. survival}} = \prod_{v=0}^{t} \exp\Big(- \underbrace{b_1 d_v^{\psi}}_{\text{hazard}} \Big)$$

$$\underbrace{d_{v+1} - d_v}_{\text{deficit accumulation}} = \mu \left[d_v - b_2 (1 + b_3 d_v)^{\alpha} h_v^{\gamma} \right]$$

Yaari (1965)
+ Dalgaard and Strulik (2014)

Endogenous
c consumption
k wealth
S cumulative survival
h health expenditures
d biological age

Exogenous
w income, R interest

Parameters
$\bar{T}$ max. lifetime; β, σ, $\bar{u}$
$b_1, b_2, b_3, \psi, \mu, \alpha, \gamma$

Green: endogenous health

Life-Cycle Model: Daily Heat Shocks and Adaptation

$$\max_{\{a_t,\, h_t,\, c_t,\, k_t\} \geq 0} \mathbb{E}_0 \sum_{t=0}^{\bar{T}} S_t\, \beta^t \underbrace{\left(\frac{c_t^{1-\sigma} - 1}{1-\sigma} + \bar{u} \right)}_{\text{annual utility}} \quad \text{s. t.}$$

$$w_t + R_{t-1}\, k_{t-1} = c_t + h_t + a_t + k_t$$

$$\underbrace{S_t}_{\text{cum. survival}} = \prod_{v=0}^{t} \prod_{\tau=1}^{365} \exp\Big(- \underbrace{\frac{b_1 d_v^{\psi}}{365}}_{\text{baseline hazard}} RR_{v,\tau} \Big)$$

$$\underbrace{d_{v+1} - d_v}_{\text{deficit accumulation}} = \mu \left[d_v - b_2 (1 + b_3 d_v)^{\alpha} h_v^{\gamma} \right]$$

$$\underbrace{RR_{v,\tau}}_{\text{relative risk}} = \exp\Big(\underbrace{g_v}_{\text{vulnerability}} \underbrace{\varepsilon_\tau^{\chi}}_{\text{heat shock}} \Big)$$

$$g_v = b_5 \underbrace{\left(\frac{d_v}{\bar{d}} \right)^{\eta}}_{\text{health}} \underbrace{e^{-\kappa a_v}}_{\text{adaptation}}$$

Endogenous
c consumption
k wealth
S cumulative survival
h health expenditures
d biological age
a private adaptation

Exogenous
w income, R interest
ε_τ daily heat shock

Parameters
$\bar{T}$ max. lifetime; β, σ, $\bar{u}$
$b_1, b_2, b_3, \psi, \mu, \alpha, \gamma$
$b_5, \eta, \kappa, \chi, \bar{d}$

Green: endogenous health

Red: heat shocks & adaptation

▸ Daily shocks
▸ Timing of events

Quantitative Implementation

Calibration

Two-step procedure

1. Impose subset of exogenous parameters (from relevant literature)

   $\sigma, \beta, r, d_0, \mu, \bar{u}, T, k_{-1}, b_1, \psi, \bar{d}$

   ▸ Table A.1

2. Estimate remaining parameters via SMM, targeting:

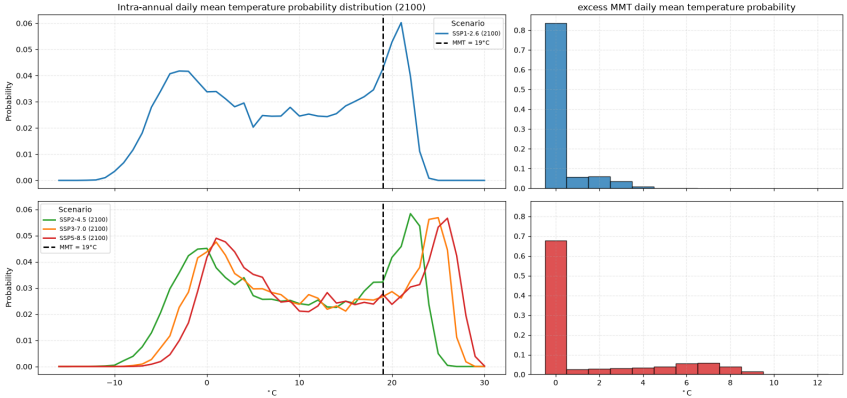
   $b_2, \gamma, b_3, \alpha, b_4, \chi, b_5, \kappa, \eta$

   - Average agent (life cycle patterns): Health expenditures, accumulated deficits, survival probabilities
   - Life expectancies across income deciles
   - Relative risk for different income groups

▸ Targets

Experimental Setup

Daily mean temperature distributions (CMIP6): SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5



Global warming 2081–2100 vs. 1850–1900 (IPCC AR6): SSP1-2.6: 1.8°C, SSP2-4.5: 2.7°C, SSP3-7.0: 3.6°C, SSP5-8.5: 4.4°C

▸ SSP1-2.6 vs. 5-8.5

Solution Strategy

Numerical solution simplified due to

- ε_τ affect LTU through S_t and do not enter the state equations
- Individual choices prior to the realization of heat shocks

$\Rightarrow$ Shocks do not induce a branching tree of future conditional states

$\Rightarrow$ Solve a simultaneous deterministic system (relaxation algorithm)

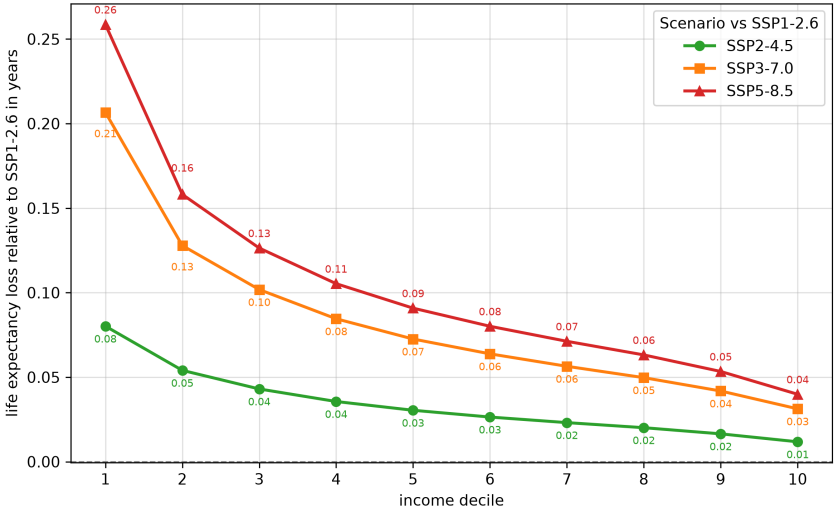
$\Rightarrow$ ... employing expected survival probabilities

$$\mathbb{E}_t(S_t) = S_{t-1}\,\mathbb{E}_t(s_t) = S_{t-1}\left(\sum_{n=1}^{N}\pi_n \exp\Big(-b_1\; d_t^{\psi}\;\frac{1}{365}\;\exp\big(g_t(a_t,d_t)\;x_n{}^{\chi}\big)\Big)\right)^{365}$$

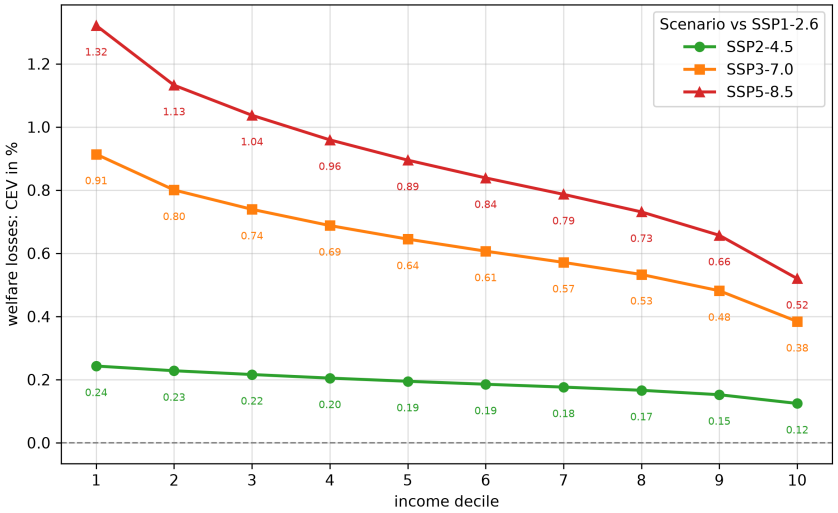
N number of bins, π_n probability of realization per bin, x_n temperature value per bin

Results

Life Expectancy

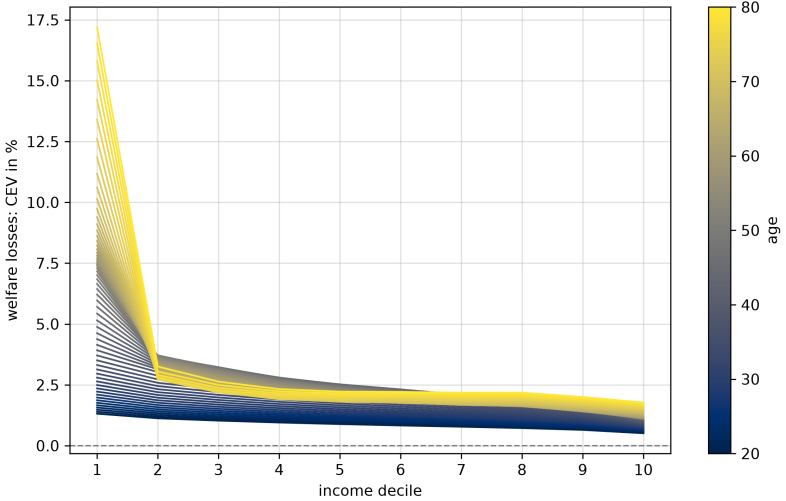


Welfare



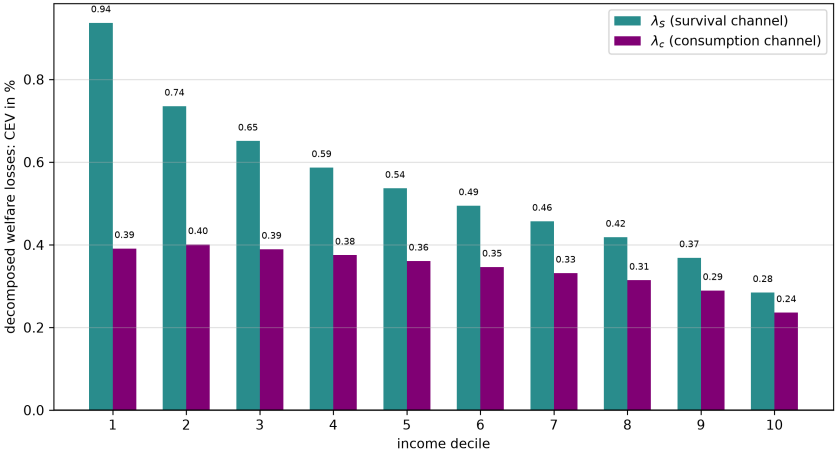
▸ Formula

Age-j-conditional Welfare SSP5-8.5 vs SSP1-2.6



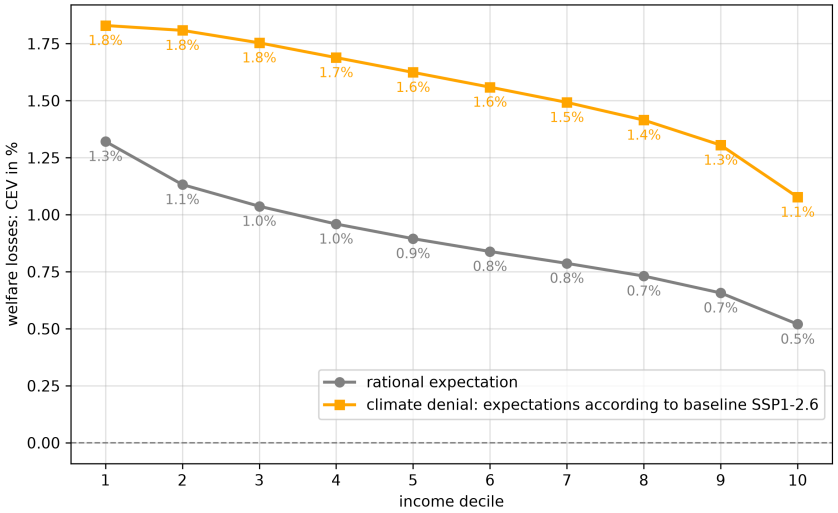
▸ Formula

Decomposition SSP5-8.5 vs SSP1-2.6



▸ Formula

Climate Denial



▸ Formula

Summary and Conclusion

Summary and Conclusion

- **Structural life-cycle model:** daily heat exposure & private abatement
- **Average agent:** $\Delta T \approx 1$ month; ΔLTU: 0.8% CEV
- **Regressive incidence:** ΔT: factor 6; ΔLTU: factor 2.5
- **Conservative damage estimates:** subnational regions, wealth ...

Thank You!

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Reduced-form Model I

Setup:

$$u(c) = \ln c + \bar{u}$$

$$s(a, d, \varepsilon) = \exp\Big(- \underbrace{b_1 d^{\psi}}_{\text{baseline hazard}} \cdot \underbrace{e^{g(a,d)\varepsilon^{\chi}}}_{\text{relative risk}} \Big)$$

$$g(a, d) = b_5 \, d^{\eta} e^{-\kappa a}$$

$$a^*(w) = \arg \max_{0 \leq a < w} \; \mathbb{E}\big[s\big(a, d(w), \varepsilon\big)\big] \, u(w - a)$$

Reduced-form Model II

Welfare:

Let $\vartheta(a,d) \equiv \mathbb{E}\, s(a,d,\varepsilon)/s_0 \in (0,1]$ and $\Phi(w) \equiv \vartheta(a^*,d)\, u(c^*)$

$$\lambda = 1 - e^{-\Lambda(w)}, \qquad \Lambda(w) = u(w) - \Phi(w) \tag{1}$$

$\Rightarrow$ The burden is regressive when $\Phi'(w) > u'(w) = 1/w$

Incidence:

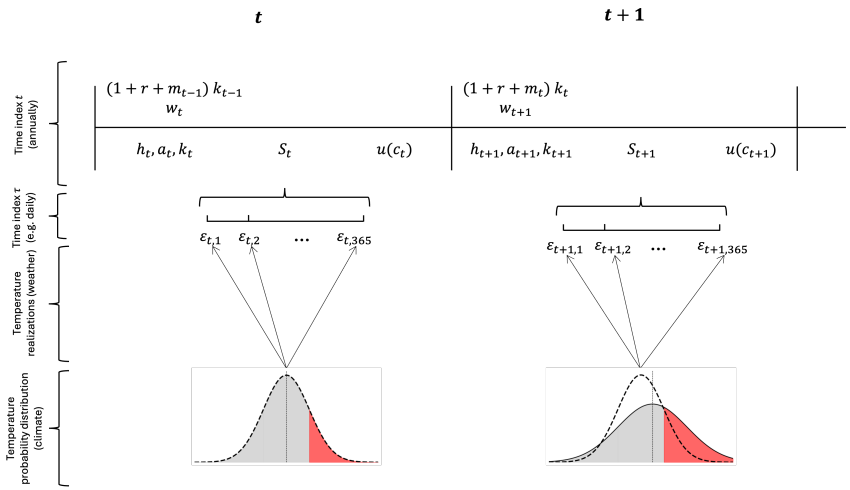
$$w\,\Lambda'(w) = \underbrace{(1-\vartheta)}_{\text{utility scale } (+)} - \underbrace{\vartheta\,\frac{a^*}{c^*}}_{\text{adaptation } (-)} - \underbrace{\nu\,\vartheta\,u(c^*)\big[\psi(H_1 - H_0) + \eta\rho H_1\big]}_{\text{biological age } (-)}.$$

Reduced-form Model III

Comparative statics:

- Adaptation share a^*/w is hump-shaped in income (zero for the poorest and small again for the rich)
- a^* rises with biological age, $da^*/dd > 0$
- A mean-preserving spread of the heat distribution raises a^*: since $e^{g\varepsilon^{\chi}}$ is convex in ε, a spread raises expected mortality and hence a^* by Jensen's inequality. It also raises residual heat mortality $1-\vartheta$, so both sides of $\frac{a^*}{w} > 1-\vartheta$ rise and the effect on w^* is ambiguous in general

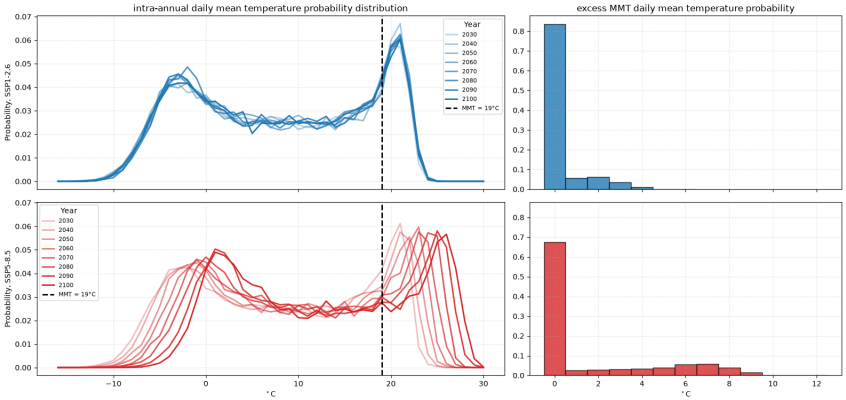
Timing of Events



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Experimental Setup

Daily mean temperature distributions (CMIP6): SSP1-2.6, SSP5-8.5



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FOCs I

$$\mathcal{L} = \mathbb{E}_0 \Bigg\{ \sum_{t=0}^{\bar{T}} \beta^t \Bigg[S_t \; u(c_t) + \lambda_t^k \left(w_t + (1 + r + m_{t-1}) k_{t-1} - c_t - h_t - k_t - a_t \right) + \lambda_t^s \left(S_t - S_{t-1} s_t \right) + \lambda_t^d \left(\mu \left(d_t - b_2 (1 + b_3 d_t)^{\alpha} h_t^{\gamma} - b_4 \right) - d_{t+1} + d_t \right) \Bigg] \Bigg\}$$

FOCs II

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 = \mathbb{E}_t(S_t)\ u_c(c_t) - \lambda_t^k$$

$$\frac{\partial \mathcal{L}}{\partial h_t} = 0 = -\lambda_t^k - \lambda_t^d\ \mu\ b_2\ \gamma\ (1 + b_3 d_t)^{\alpha} h_t^{\gamma - 1}$$

$$\frac{\partial \mathcal{L}}{\partial a_t} = 0 = -\lambda_t^k - \lambda_t^s\ S_{t-1}\ \mathbb{E}_t(s_a(a_t, d_t)) + \beta\ \lambda_{t+1}^k\ \mathbb{E}_t(m_a(a_t, d_t)\ k_t)$$

$$\frac{\partial \mathcal{L}}{\partial k_t} = 0 = -\lambda_t^k + \beta\ \lambda_{t+1}^k (1 + r + \mathbb{E}_t(m_t))$$

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial d_{t+1}} = 0 = &-\lambda_t^d + \beta \lambda_{t+1}^d \left(\mu \left[1 - b_2 \alpha b_3 (1 + b_3 d_{t+1})^{\alpha - 1} h_{t+1}^{\gamma}\right] + 1\right) \\ &- \beta \lambda_{t+1}^s S_t\, \mathbb{E}_t[s_d(d_{t+1}, a_{t+1})] \\ &+ \beta^2 \lambda_{t+2}^k\ k_{t+1}\ \mathbb{E}_t[m_d(d_{t+1}, a_{t+1})]\end{aligned}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbb{E}_t(S_t)} = 0 = u(c_t) + \lambda_t^s - \beta\ \lambda_{t+1}^s \mathbb{E}_t(s_{t+1})$$

Daily Heat Shocks and Annual Survival

Annual conditional survival:

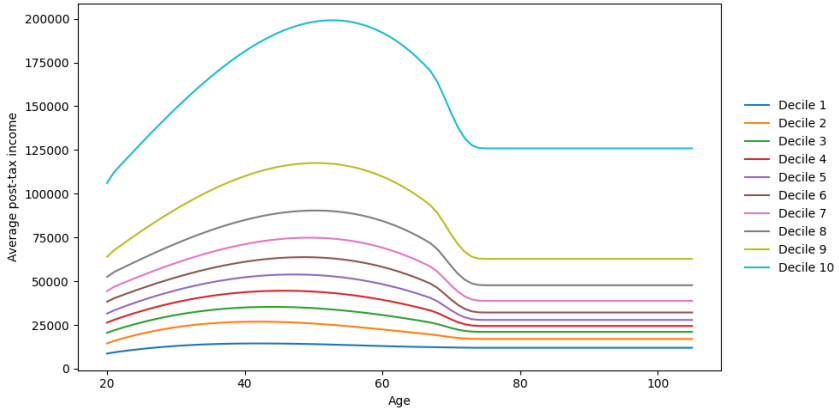
$$\ln s_t = \ln \prod_{\tau=1}^{365} e^{-\frac{b_1 d_t^{\psi}}{365} RR_{t,\tau}} = - \underbrace{b_1 d_t^{\psi}}_{\text{annual baseline hazard}} \cdot \underbrace{\frac{1}{365} \sum_{\tau=1}^{365} RR_{t,\tau}}_{\text{average relative risk}}$$

$$RR_{t,\tau} = e^{g_t \varepsilon_{t,\tau}^{\chi}}, \qquad g_t = b_5 \left(\frac{d_t}{\bar{d}} \right)^{\eta} e^{-\kappa a_t}, \qquad S_t = \prod_{v=0}^{t} s_v$$

- **Mixed frequency:** daily shocks τ, annual decisions t
- **Jensen:** convex damages $\Rightarrow$ annual mean temperature understates mortality
- **Heat only:** $\varepsilon_{t,\tau} = \max(T_{t,\tau} - \text{MMT}, 0)$, MMT exogenous
- **Vulnerability** g_t: health (d_t) and adaptation (a_t)

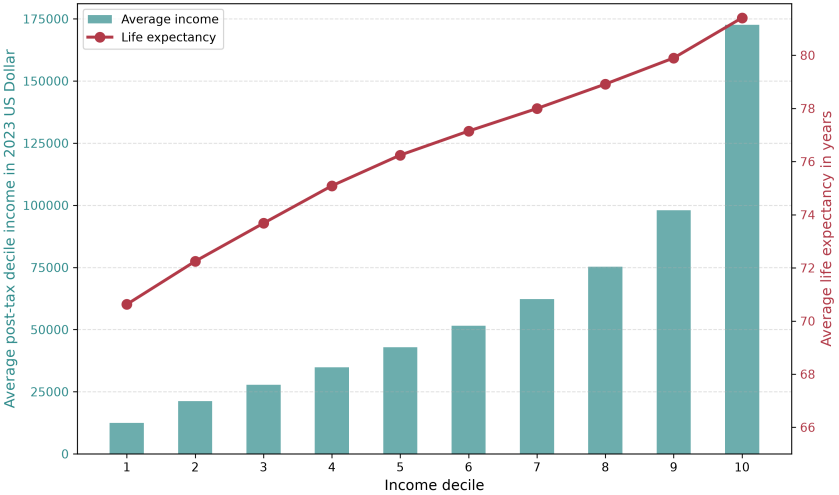
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Life-cycle Income Profiles



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Income and Life Expectancy



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Calibration: Parameter Values

A1: Set outside of estimation

Param.	Value	Source
σ	1.01	Strulik and Grossmann (2025)
β	0.94	Strulik and Grossmann (2025)
r	0.06	Strulik and Grossmann (2025)
d_0	0.036	Strulik and Grossmann (2025)
μ	0.034	Strulik and Grossmann (2025)
$\bar{u}$	-0.2	VSL of \$10m at age 40 (Hall and Jones, 2007)
$\bar{T}$	90	Max. lifespan 110 years
k_{-1}	0	Assumption
b_1	3.3842	Fitted: association $d \leftrightarrow S$
ψ	2.5629	Fitted: association $d \leftrightarrow S$
$\bar{d}$	d_0	Normalization

SMM with block-size-scaled log deviations; multiple initial guesses around first-stage estimates.

A2: Estimated by SMM ($\hat{\theta}_N$)

Param.	Value
b_2	0.000807
γ	0.2505
b_3	0.01459
α	488.09
b_4	0.0000
χ	1.989
b_5	0.000953
κ	0.001178
η	1.304

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Calibration: Targets (ζ_N^{data})

Life-cycle profiles, average agent

Age	Health exp. ($/year)	Deficits d	Survival S
20	1,284.1	0.036	0.986
30	2,096.5	0.047	0.973
40	3,143.3	0.062	0.957
50	3,863.3	0.082	0.923
60	4,413.4	0.111	0.851
70	5,240.6	0.152	0.718
75	5,696.2	–	–
80	–	0.209	0.478
90	–	0.288	0.145
100	–	0.400	0.006

Heat-mortality relative risk

Agent	Scenario	Age	Temp.	RR
Average	SSP1-2.6	40	MMT+5	1.05
Average	SSP1-2.6	85	MMT+5	1.10
Average	SSP1-2.6	40	MMT+7	1.10
Average	SSP1-2.6	85	MMT+7	1.20
Decile 9	SSP1-2.6	85	MMT+7	1.09
Decile 2	SSP1-2.6	85	MMT+7	1.61
Average	SSP5-8.5	85	MMT+7	1.16

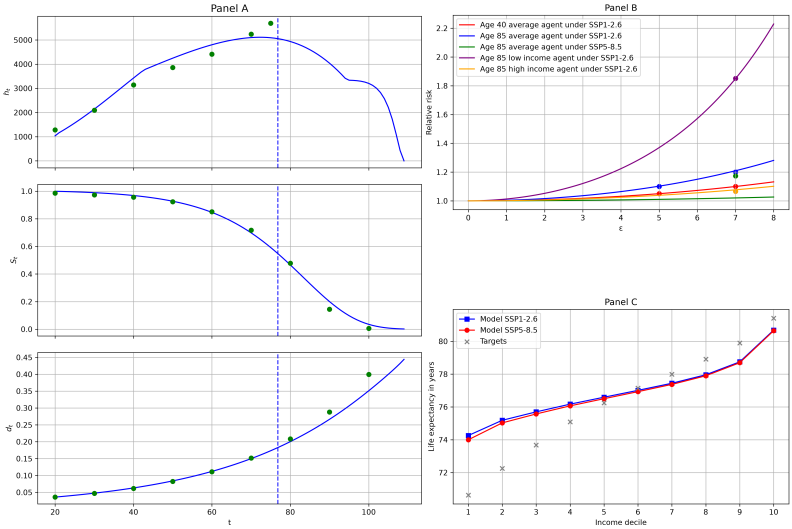
Life expectancy (years) by income decile

Decile	1	2	3	4	5	6	7	8	9	10
LE	70.19	71.81	73.24	74.65	75.81	76.71	77.56	78.48	79.46	80.97

MMT: minimum mortality temperature. –: not targeted.

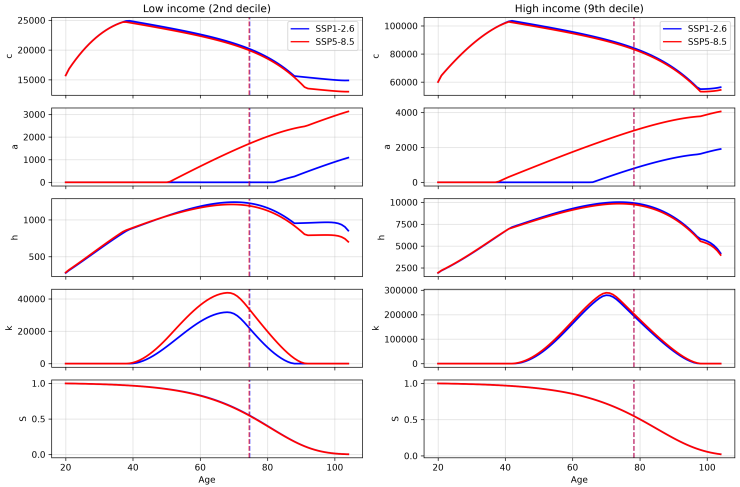
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Calibration Outcome: Model Fit



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Life-Cycle Trajectories



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CEV Formulas

The CEV indicates the permanent proportional change in the SSP1-2.6 benchmark consumption stream required to equalize expected lifetime utility across heat scenarios $Z \in \{4.5, 7.0, 8.5\}$:

$$\mathbb{E}_0\left[\sum_{t=0}^{\bar{T}} \beta^t S_t^{2.6} u\left((1-\lambda)c_t^{2.6}\right)\right] = \mathbb{E}_0\left[\sum_{t=0}^{\bar{T}} \beta^t S_t^{Z} u\left(c_t^{Z}\right)\right]$$

$$\mathbb{E}_j\left[\sum_{t=j}^{\bar{T}} \beta^{t-j} \frac{S_t^{2.6}}{S_j^{2.6}} u\left((1-\lambda_j)\, c_t^{2.6}\right)\right] = \mathbb{E}_j\left[\sum_{t=j}^{\bar{T}} \beta^{t-j} \frac{S_t^{Z}}{S_j^{Z}} u\left(c_t^{Z}\right)\right].$$

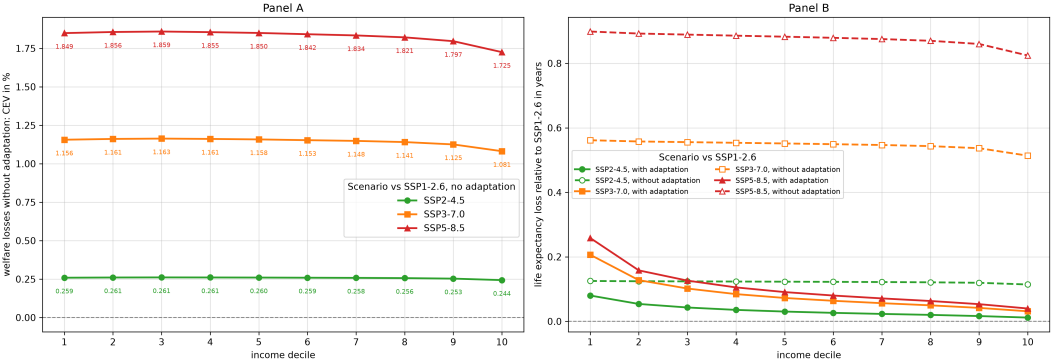
$$\mathbb{E}_0\left[\sum_{t=0}^{\bar{T}} \beta^t S_t^{2.6} u\left((1-\lambda_S)c_t^{2.6}\right)\right] = \mathbb{E}_0\left[\sum_{t=0}^{\bar{T}} \beta^t S_t^{Z} u\left(c_t^{2.6}\right)\right]$$

$$\mathbb{E}_0\left[\sum_{t=0}^{\bar{T}} \beta^t S_t^{2.6} u\left((1-\lambda_c)c_t^{2.6}\right)\right] = \mathbb{E}_0\left[\sum_{t=0}^{\bar{T}} \beta^t S_t^{2.6} u\left(c_t^{Z}\right)\right]$$

$$\mathbb{E}_0\left[\sum_{t=0}^{\bar{T}} \beta^t\, S_t^{2.6}\, u\left((1-\lambda^*)\, c_t^{2.6}\right)\right] = \mathbb{E}_0\left[\sum_{t=0}^{\bar{T}} \beta^t\, S_t(a_t^{2.6}, h_t^{2.6}, \varepsilon^{8.5})\, u\left(c_t^{2.6}\right)\right]$$

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Results Absent Adaptation



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